

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Branch: Civil Engineering

Semester: VIII

Subject Code & Name: BTCVSS801D & Maintenance & Repair of concrete Structure

Max Marks: 60

Date:15-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

12

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| A) What are the reasons for failure of repairs? | CO1 | 6 |
| B) Enlist different types of corrosions in concrete structures? Explain Carbonation induced corrosion in detail. | CO1 | 6 |
| C) What is the necessity of corrosion management? | CO1 | 6 |

Q.2 Solve Any Two of the following.

12

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|---|-----|----------|
| A) Explain in brief Corrosion induced cracking and spalling in building. | CO2 | 6 |
| B) State and explain different sources of sulphate which can attack on strength and durability of building structure. | CO2 | 6 |
| C) What is DEF? Explain the mechanism of DEF induced cracking. | CO2 | 6 |

Q. 3 Solve Any Two of the following.

12

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| A) Explain in brief frost attack. Suggest suitable measures to reduce effect of frost attack. | CO3 | 6 |
| B) Explain in detail with the help of suitable sketches, crack patterns developed due to ASR. | CO3 | 6 |
| C) What are the different causes of deterioration of cementitious system? | CO3 | 6 |

Q.4 Solve Any Two of the following.

12

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|---|-----|----------|
| A) Explain the different steps of assessment of condition of RCC structures. | CO4 | 6 |
| B) What is ground penetrating radar system? How it locate the delaminated concrete? | CO4 | 6 |
| C) How the concrete surface repair is achieved? | CO4 | 6 |

Q. 5 Solve Any Two of the following.

12

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| A) State the different questions needs to be addressed during repair analysis? | CO5 | 6 |
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| B) Explain the process of strengthening of slab and beams. | CO5 | 6 |
| C) State the different methods of strengthening of column and explain any one in detail. | CO5 | 6 |

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